Fab26 – Reimagine the Future. Fab26 Research Papers Stream

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Abstract

FAB26 marked the 25th anniversary of the Fab Lab phenomenon, its network and movement. The conference was as much looking back at that legacy as looking forward to the next 25 years. The fifteen papers in this year's research stream stand at that watershed, building on the past and for the future. The papers fall into three clusters, presented across three sessions: Teaching & Learning, Making with Communities, and Sustaining and Scaling.

Teaching & Learning: From First Stitch to Education 5.0 — pedagogy, tools, and AI across the learner's journey — opens with a small piece of hardware: Leirado et al. develop sew:bit, an open-source expansion board that makes the BBC micro:bit work for e-textiles much as the LilyPad once did for Arduino. LaPrade proposes The Armature, a community-first framework carrying middle-school makers at a Title I school in Cleveland toward a professional CAD credential. Yalçın reports seven years of a K-12 FabLab in Istanbul in which students become co-authors of peer-reviewed HCI research. Llàcer Caro and Muñoz Zanón ask what critical thinking means once generative AI is ubiquitous, and offer a toolkit built on deliberate friction. Hernández-Montoya and Miranda close the session with an AI-mediated model joining remote, physical, and mobile Fab Labs at a Costa Rican distance university.

Making with Communities: From Visiting to Becoming Infrastructure — resilience and sustainability transitions on the ground — moves outward. Liu and Liao use a seven-day Fab City Challenge to read Shenzhen's Nantou Ancient City through fabrication. Espinoza Gómez et al. ask honestly what it would take to evaluate an itinerant digital fabrication programme in rural Peru, having first run one without a baseline. de Castro Lacerda and Tenório Ramalho de Abreu describe a participatory method for textile waste with women upcycling decommissioned bus-driver uniforms in Brasília. Regina et al. explore village-scale green hydrogen in coastal Bali through five co-designed interventions, built on the existing governance of the Balinese *banjar*. Anaya Hernández asks whether maker networks actually hold the data needed to be activated in a disaster, and finds the answer uneven.

Sustaining and Scaling: From One Workshop to the Full Stack — governance, stakeholders, and measurement of fab initiatives — turns to what makes any of this last. Stadlmann and Holzinger treat maker workshops for children as multi-stakeholder services and find that stakeholders do not agree on what those services are for. Doyo draws on ten years of a Japanese university FabLab to argue that openness is a three-layer achievement of place, governance, and participatory mediation. León Lucano proposes Fab Lab venture studios as a university investment framework for Peru. Freundt et al. report a national faculty-development programme reaching 27 public technical institutes. Vivanco closes with the Fab City Full Stack Metrics Framework, an attempt to make local activity measurable without severing it from the systems it sits in.

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Twenty-nine papers were submitted to this year's stream. The stream's policy is double-blind peer review, with each paper receiving at least two reviews; on that basis fourteen were accepted and one was rerouted as a preprint. Their thirty-one authors work in twenty-two institutions across thirteen countries. Each paper is deposited separately and carries its own DOI, listed in the contents that follow; this volume gathers their abstracts and is the citable record of the stream as a whole. The research papers stream has run since Fab8 in Wellington in 2012, a history set out in Troxler, *Fab Lab Research Papers. From Experiment to Expression* (10.5281/zenodo.1344101); the most recent volume was *Fab25 Czechia – Bridge the Gap* (10.5281/zenodo.16789442).

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Keywords

Fab Lab, Digital Fabrication, Impact, Education, Community, Governance

1 TEACHING & LEARNING: FROM FIRST STITCH TO EDUCATION 5.0

Pedagogy, tools, and AI across the learner's journey

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DOI: 10.5281/zenodo.21712564

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Melvin LaPrade
DOI: 10.5281/zenodo.21712566

1.3 From Makers to Researchers: A Model for Student-Driven HCI Research in a K-12 FabLab
Sedat Yalçın
DOI: 10.5281/zenodo.21712570

1.4 Friction First: A Toolkit for Teaching Critical Thinking in an AI Saturated Classroom
Sofia Llàcer Caro and Jorge Muñoz Zanón
DOI: 10.5281/zenodo.21712572

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Diana Hernández-Montoya and Jhonattan Miranda
DOI: 10.5281/zenodo.21712576

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3 SUSTAINING AND SCALING: FROM ONE WORKSHOP TO THE FULL STACK

Governance, stakeholders, and measurement of fab initiatives

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DOI: 10.5281/zenodo.21712593

3.2 Reconfiguring Openness: Boundary Reconfiguration and Institutional Scaling in a Japanese University FabLab
Daisuke Doyo
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Jair Jesús León Lucano
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3.4 Scaling maker-based pedagogies in public technical education: A decentralized faculty development model through Peru's Institutes of Excellence (IDEX)
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3.5 Fab City Full Stack Metrics Framework: An Actionable Methodology for Multi-Scalar Implementation
Tomas Vivanco
DOI: 10.5281/zenodo.21712601

1 Teaching & Learning: From First Stitch to Education 5.0

Pedagogy, tools, and AI across the learner's journey

1.1 Sew:bit: An Open-Source Expansion Board Extending the micro:bit for E-Textile and Wearable Computing in K-6 Education

Julia Leirado, Xavier Domínguez and Santiago Fuentemilla

E-textiles are increasingly integrated in K–6 physical computing education, but existing platforms often remain disconnected from widely adopted classroom ecosystems such as the BBC micro:bit. Boards like LilyPad, Flora, and Circuit Playground have shaped wearable computing practices, yet there is limited hardware specifically designed to adapt the BBC micro:bit, the most widely used physical computing platform in K-6 education, for textile-based making.

We developed sew:bit to address this gap. Sew:bit is an open-source expansion board for the BBC micro:bit designed within a Fab Lab context to support wearable and e-textile practices in K-6 education. Building on a comparative analysis of existing e-textile platforms, the project examines how form factor, connectivity, and fabrication constraints shape classroom use. This paper presents the iterative design and digital fabrication process of sew:bit from early conceptual sketches and PCB layout decisions to prototyping and the development of a beta kit enabling introductory textile-based computing artifacts.

Field deployments were conducted with three distinct participant groups: kids, teachers, and members of the maker community. These deployments provided opportunities to combine observational insights with structured feedback, generating iterative input on usability, accessibility, and pedagogical integration. Across these contexts, early findings suggest that adapting the BBC micro:bit for textile-based making lowers barriers to entry and broadens participation in physical computing. The project contributes a design case study that connects open-source hardware development, digital fabrication practices, and textile-based computing within K-6 education and Fab Lab ecosystems.

Keywords: micro:bit, E-textiles, K-6, Open hardware, Physical computing education, Digital Fabrication, Computational Making

DOI: 10.5281/zenodo.21712564

1.2 What the Turtle Still Needs: From Papert's Promise to The Armature — A Community-Driven CSWA Pathway for Middle School Makers

Melvin LaPrade

This conceptual and position paper argues that the Fab Lab movement has scaled access to digital fabrication more successfully than it has scaled a durable, community-accountable implementation architecture for youth in under-resourced urban communities — the students Mel King's founding vision for community Fab Labs most urgently sought to serve. It introduces The Armature, a four-strategy community-first framework — Community-Anchored Design, Identity-Forward Pedagogy, Spiral Credential Mapping, and Documented Iteration as Assessment — grounded in Constructionism 3.1, Objects-to-Become-With, and Fugitive Pedagogy, and extended through TPACK+C+CP, an original contribution of this paper. The Scholar-Architect Maker Academy at Kenneth Clement Boys Leadership Academy (KCBLA), a Title I school in Cleveland, Ohio, serves as an illustrative design case for a Grade 7–8 pathway toward Certified SOLIDWORKS Associate (CSWA) examination readiness in Grade 9, supported by the first documented alignment of CSWA competencies to Ohio Learning Standards and a backward-mapped Grade 6–9 scope and sequence. As Ohio House Bill 66 mandates middle school Career and Technical Education (CTE) amid a documented absence of causal evidence, the paper reframes that gap as a design problem and positions The Armature as one community-first architecture capable of guiding Title I implementation before evidence can accumulate.

Keywords: Digital fabrication, middle school CTE, CSWA credential, constructionism, fugitive pedagogy, The Armature, community partnership, Title I education

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1.3 From Makers to Researchers: A Model for Student-Driven HCI Research in a K-12 FabLab

Sedat Yalçın

This paper presents a seven-year longitudinal account of a K-12 FabLab functioning not only as a fabrication space but as a continuous research environment in which high school students move from makers to co-designers and co-authors of peer-reviewed publications. At Hisar School ideaLab, students have developed, evaluated, and published interactive systems in venues including ACM HRI, AIED, HCII, ACM MUM, ACM SCF, IDC, and Constructionism. Drawing on six published systems spanning robotics, AI education, augmented reality, parametric design, and tangible interfaces, the paper examines the structural conditions that enabled student research participation across design, fabrication, evaluation, and writing. Across these systems, over 300 K-12 learners participated in structured evaluations, while high school students in the lab contributed as emerging researchers. The paper identifies three enabling structural conditions: a coordinator bridging pedagogy and research, a lab infrastructure collapsing the distance between prototyping and deployment, and student participation structured toward co-authorship from the outset. Rather than presenting the projects as isolated outcomes, the paper proposes a practical model for FabLab communities seeking to support student-driven HCI research within educational settings.

Keywords: FabLab, student-driven research, HCI education, K-12, co-design, digital fabrication, AI education, iterative design, tangible interfaces

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1.4 Friction First: A Toolkit for Teaching Critical Thinking in an AI Saturated Classroom

Sofia Llàcer Caro and Jorge Muñoz Zanón

This paper takes the position that the mass adoption of LLMs and multimodal AI systems in educational contexts is unlike other technologies and, above all, an epistemic shift. We believe that the most powerful thing we can do in response is to focus on building critical thinking in the classroom.

For this, we propose an educational framework built around critical inquiry and technological sovereignty. The goal is for students to learn to question and understand AI, not just blindly use it. The framework provides five design principles and a cyclical model of critical thinking development which different educational communities can adapt to their own contexts and needs.

Drawing from the authors' teaching experiences, two case studies are presented to examine the framework in practice: a vocational training program in mechanical fabrication, and a digital skills course for adults with disabilities. The paper concludes with a proposal for a collective pedagogical playbook: an open, living resource where educators from Fab Labs, universities, and informal learning contexts can contribute their own methodologies, build upon others' and document the process. The proposals in this paper are starting points, not rules, and the authors believe they should be co-created with the education community.

Keywords: Multimodal AI, Education, Technological Sovereignty, Pedagogical Friction, Critical Inquiry

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1.5 An Integrated AI-Mediated Model for Remote, Physical, and Mobile Fab Labs in the Transition to Industry 5.0

Diana Hernández-Montoya and Jhonattan Miranda

Industry 5.0 has shifted the benchmark for innovation from digital efficiency alone to human-centered, sustainable, and resilient socio-technical systems, yet Fab Lab research still offers limited guidance on how artificial intelligence can support distributed learning through making without reducing design work to generic content generation. This article reports a design-based, embedded mixed-methods case study of Fab Lab Kä Träre at Universidad Estatal a Distancia in Costa Rica. The study reconstructs the institutional problem that motivated the intervention, analyses the design and operation of an integrated remote, physical, and mobile Fab Lab ecosystem, and incorporates bounded pedagogical validation from institutional materials and a complementary survey administered in an associated AI-mediated educational activity. The intervention implemented AI-mediated guidance for problem framing, personalization, and viability support while preserving human judgment, mentoring, and the material conditions of making. The results show positive perceived educational value, particularly in the theory-to-practice connection, learning through analysis and design, and the potential to incorporate similar activities into other courses. The article argues that the contribution of AI in this case lies in the deliberate design of an integrated environment that reduces avoidable uncertainty, widens access, and links remote guidance, embodied fabrication, and territorial reach.

Keywords: Fab Lab, Artificial Intelligence, human-centered innovation, Education 5.0, 21st-century skills

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2 Making with Communities: From Visiting to Becoming Infrastructure

Resilience and sustainability transitions on the ground

2.1 Living Layers: Teaching, Designing, and Reimagining Nantou through Fab City Challenge

Jingqi Kay Liu and Lit Liao

This paper examines how fabrication-based educational programs can support students in engaging with complex urban environments through situated inquiry, stakeholder engagement, and prototyping. Using the 2025 Fab City Challenge (FCC) in Nantou Ancient City in Shenzhen as a case study, it asks: How can a short-term fabrication-based educational program support students in identifying, interpreting, and responding to the layered social and spatial conditions of a city in transition?

Over seven days, participating high school students engaged in site observation, spatial mapping, stakeholder interviews, digital fabrication workshops, and iterative prototyping. The curriculum was structured around four phases—immersion, framing, intervention, and sharing—drawing upon principles of situated learning, inquiry-based education, and research-through-design.

Analysis of the process and resulting projects reveals four key findings: (1) visibility significantly shapes patterns of urban participation; (2) heritage is experienced through everyday traces as much as through official preservation narratives; (3) mapping can reveal hidden relationships between circulation, access, and stakeholder perspectives; and (4) digital fabrication functions not only as a production tool but also as a method of urban inquiry.

The paper contributes a pedagogical framework for fabrication-based urban learning and demonstrates how making can serve as a means of investigating, interpreting, and engaging with cities in transition.

Keywords: urban resilience, situated learning, design pedagogy, urban transformation, sustainability transition

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2.2 Designing an Impact Evaluation Framework for Itinerant Digital Fabrication Interventions and technology transfer in Rural Peru

Silvana Espinoza Gómez, George Fajardo Soria and Benito Juárez Vélez

Many rural communities in Madre de Dios face limited opportunities to develop alternative livelihoods. This structural constraint often perpetuates long standing cycles of dependence on activities such as illegal mining or logging, limiting social mobility and reinforcing restricted economic pathways. In response to this challenge, inclusive technology interventions were implemented in Madre de Dios by FabLab Perú using an itinerant digital fabrication model centered on practical skill development and locally relevant problem solving, following the “Symbiocreation” methodology proposed by FabLab Perú.

This paper analyzes the first intervention as a formative implementation experience. Although no baseline data, structured outcome measures, or randomized assignment were collected in this initial phase, the experience provides critical insights to inform the initial stages of a future impact evaluation design. In the paper we analize different frameworks of impact evaluation, based on them we identify key assumptions, outcome gaps, and operational constraints that limited evaluability in the pilot phase.

Building on these lessons, the paper outlines methodological considerations for possible new interventions under the same itinerant model. These include clarifying research questions, identifying measurable intermediate outcomes such as skills acquisition and technological agency, and exploring potential indicators related to engagement in alternative productive activities. The paper also discusses how a credible counterfactual might be constructed and

what ethical constraints must guide these decisions.

Rather than presenting a finalized evaluation design, this work establishes the conceptual and methodological foundations necessary to initiate a rigorous and context sensitive impact measurement process in future implementations.

Keywords: Rural, Co-design, Social Impact Measurement, Itinerant Fab Lab, Educational Technology, Industry 4.0 Education

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2.3 Decolonizing Circularity: Community Validation of a Social Technology for Textile Waste Mitigation in the Global South

Rafaella de Castro Lacerda and Breno Tenório Ramalho de Abreu

Circularity has become a central paradigm in global sustainability agendas, yet dominant frameworks often assume infrastructures, traceability systems and recycling capacities that are unevenly available across the Global South. This paper presents the Hybrid Monster Mitigation Methodology (HMMM), a participatory social technology for diagnosing textile incompatibilities and supporting community-based circular design. Grounded in Cradle to Cradle, decolonial theory and participatory action research, the study was validated with women engaged in a community fashion upcycling programme using decommissioned bus-driver uniforms. The paper details participant recruitment, data analysis, workshop observations, material exercises and public outcomes. Data included facilitator field notes, process documentation, photographs, project diaries, compatibility tables and collective discussions. Findings show that material literacy can be developed through accessible practices such as fibre identification, compatibility mapping and redesign decisions oriented toward future disassembly. Beyond technical results, the public presentation of five complete looks demonstrated how textile waste can be resignified as a territorial resource and as a platform for authorship. The paper argues that Fab Labs can integrate HMMM as a pre-fabrication material literacy layer for situated circular transitions.

Keywords: textile waste, circular economy, Cradle to Cradle, decolonial design, Fab Labs

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2.4 Desa Hidrogen Hijau: A Dual-Track Methodology for Village-Scale Energy Transition Integrating Social and Technical Development in Coastal Bali

Elaine Regina, Tafia Sabila, Tomas Diez, Wan Zaleha Radzi and Athina Ibrahim

Energy-transition research lacks methods to integrate technical development with community social and cultural realities. Technology-first interventions in the Global South often fail socially, yet literature offers no tested model for structuring recursive exchanges between community insights and engineering. This paper tests a dual-track methodology—parallel Community and Technology Development workstreams coupled by structured feedback loops—within a village-scale "sandbox." Here, energy systems are co-developed with residents, academic, enterprise, and government actors, orchestrated by an embedded fab lab. Evaluating a two-year (2023–2025) participatory action-research study of the Desa Hidrogen Hijau (Green Hydrogen Village) project in Serangan, Bali, we ground the work in multiscalar village reading, commons governance, and appropriate-technology principles. Across five co-developed interventions, parallel development surfaced design parameters absent from prior specifications. We demonstrate that the Balinese banjar, anchored by the bale banjar, functions as both governance infrastructure and a replicable substrate, proving energy transition can be built on existing culture, not imposed as standalone hardware. Finally, we document an enthusiasm-to-maintenance gap, highlighting the need for a third track of institutional embedding.

Keywords: dual-track methodology, participatory action research, village-scale energy transition, commons governance, appropriate technology

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2.5 Assessing Data Availability across Maker Networks: Measuring Information Preparedness for Disaster Response

Antonio de Jesús Anaya Hernández

While the COVID-19 pandemic demonstrated the capacity of global maker networks to act as critical nodes in emergency supply chains, there remains a lack of formal assessment regarding their information readiness for future disaster response interventions. This study measures what data currently exists across maker ecosystem platforms and what would be required to activate them during a crisis. It performs a quantitative and geospatial analysis on ten online maker platforms compiled between 2021 and 2026 through the Internet of Production Alliance's Open-Know-Where (OKW) initiative, harmonized through a 20-class semantic taxonomy and evaluated using semantic intersection analysis and hierarchical clustering. The findings reveal a hard boundary dividing the sources into two tiers: operationally rich platforms detailing manufacturing processes, equipment, and materials, and directory-level sources providing only location registration — a systemic vulnerability, as the network's manufacturing intelligence depends on a small, fragile subset of sources. The study introduces the Distributed Manufacturing Data Kit (DMDK), an open pipeline for extracting, harmonizing, and visualising manufacturing data across incompatible platforms, and grounds the proposed Disaster Response Preparedness Index (DRPI) in the identified gaps, defining the data records that maker networks, NGOs, and policymakers should collect.

Keywords: distributed manufacturing, disaster response, data interoperability, maker networks, open data

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3 Sustaining and Scaling: From One Workshop to the Full Stack

Governance, stakeholders, and measurement of fab initiatives

3.1 Designing Children's Maker Workshops as Multi-Stakeholder Services: A Service-Marketing Perspective (Preprint)

Christian Stadlmann and Andrea Holzinger

Children's makerspace programmes often start as funded pilot projects and then face the challenge of becoming sustainable services. Against this background, this study examines children's maker workshops from a service-marketing perspective and conceptualizes them as multi-stakeholder services embedded in a maker ecosystem. Building on service-dominant logic, the study empirically analyzes the expectations of key stakeholder groups—enablers, financiers, and facilitators—across two distinct workshop formats within an Austrian maker initiative. A mixed-methods design was applied. In an ex-ante survey, trainers, parents, teachers/school leaders, and company representatives rated and ranked literature-derived expectation items of maker workshops to reveal relative priorities. Results were subsequently validated and extended in several co-creation workshops. Findings show that stakeholders see many workshop design goals as important, especially hands-on skills, safe tool use, self-confidence, teamwork, and enjoyment, while digital activities, the presentation of workshop outcomes, creativity elements, and a sense of group belonging are more contested. However, the prioritization of workshop design elements is not convergent among stakeholders and require hence, an adaptation to the context of the different stakeholder groups. Qualitative results highlight additional ecosystem-level expectations (e.g., alignment with schools, structured school–company collaboration, inclusion, and organisational reliability), suggesting that long-term value requires an orchestration of the maker ecosystem beyond individual workshops. The paper provides actionable design principles and a transferable expectation-based instrument for practitioners, makers, and researchers.

Keywords: K-8 makerspace education, multi-stakeholder expectations, service-dominant logic, service ecosystem

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3.2 Reconfiguring Openness: Boundary Reconfiguration and Institutional Scaling in a Japanese University FabLab

Daisuke Doyo

University-based FabLabs are frequently described as open access spaces aligned with the Fab Charter, yet within universities, openness is shaped by safety regulations, liability frameworks, and institutional governance. This paper examines how openness is operationalized over time through a ten-year longitudinal study (2015–2025) of a university FabLab in Japan that transitioned from a suburban campus facility to an urban-facing fabrication space within a publicly accessible co-creation environment. Drawing on boundary work theory, institutional work, and commons governance, the study proposes a three-layer boundary reconfiguration model encompassing spatial embedding, institutional codification, and participatory mediation. Participation records show external users growing from 11–23% to approximately 28% after relocation and governance redesign, alongside total users increasing from roughly 1,100 to over 3,300. The findings demonstrate that spatial visibility alone does not produce openness; durable openness emerges only when spatial, institutional, and participatory dimensions align. The paper reframes openness as a governance achievement rather than a normative attribute, offering a processual model for how university FabLabs can function as civic infrastructures within regulated environments.

Keywords: University FabLab, Openness, Boundary Reconfiguration, Civic Infrastructure, Institutional Governance

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3.3 Institutionalizing technological sovereignty through fab lab venture studios: A university investment framework for intellectual capital development.

Jair Jesús León Lucano

Universities in emerging economies frequently generate early-stage technological knowledge but lack institutional mechanisms to retain, scale, and valorize the intellectual capital produced within academic innovation ecosystems. This study explores an institutional framework that integrates Fab Lab Venture Studios (FLVS) with an Academic Corporate Venture Capital (ACVC) model embedded within a university governance structure. Using the 2025 Fellowship Program as a technical pilot implemented across four Peruvian cities (Lima, Arequipa, Cusco, and Huancayo), the research analyzes how distributed prototyping infrastructures can support venture creation and localized technological capability development. The study combines innovation ecosystem indicators with an experimental "learning-by-investing" approach involving 24 university-linked projects connected to Business-to-Business (B2B) and Business-to-Government (B2G) actors. Findings suggest that institutional investment governance, media visibility, and structured intellectual property management play a critical role in improving the transition from prototyping to early-stage commercialization. The paper further proposes a dual valuation approach based on venture capital logic and cost-to-duplicate estimation to support internal asset recognition in early-stage hardware-oriented projects. Although the study is limited to an exploratory pilot, the framework offers potential applicability for universities operating in resource-constrained innovation ecosystems across Latin America and other emerging economies.

Keywords: Technological sovereignty, intellectual capital valorization, biopolitical autonomy, fab Lab

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3.4 Scaling maker-based pedagogies in public technical education: A decentralized faculty development model through Peru's Institutes of Excellence (IDEX)

Victor Freundt, Cesar Lopez, Luzmila Rojas and Tito Portocarrero

As digital fabrication becomes increasingly accessible, technical education faces the challenge of developing educators' capacities to integrate maker-based pedagogies rather than simply providing technological infrastructure. This paper examines the implementation and immediate outcomes of a national faculty development program delivered through Peru's Institutes of Excellence (IDEX), a network of 27 public technical education institutions.

Implemented in collaboration with the Ministry of Education, the program combined 15 synchronous virtual sessions with three regional two-day Fab Lab workshops, engaging 195 faculty members online and 151 in practical training. The study employed an embedded mixed-methods program evaluation using attendance records, structured learning activities, project documentation, standardized rubrics, satisfaction surveys, and facilitator observations. Quantitative data were analyzed using descriptive statistics, while qualitative evidence supported a comparative analysis of the 27 institutional innovation projects developed during the program.

Results showed high participation and engagement, with a 77.4% attendance rate for the workshops, a mean academic score of 16.47/20, and high participant satisfaction. Three recurring innovation patterns emerged—technology-driven, context-driven, and hybrid approaches—while common challenges included usability, system integration, and scalability. The findings demonstrate the feasibility of a standardized, decentralized faculty development model and provide a scalable framework for strengthening innovation capabilities through

maker-based education in technical and vocational systems.

Keywords: maker education, fab labs, project-based learning, faculty development, technical education

DOI: 10.5281/zenodo.21712599

3.5 Fab City Full Stack Metrics Framework: An Actionable Methodology for Multi-Scalar Implementation

Tomas Vivanco

Implementing impact metrics in Fab Labs and the Fab City network requires more than a list of indicators. It requires a coherent structure that links local activity with wider urban, regional, ecological, and planetary systems. The Fab City Full Stack Metrics Framework provides that structure through seven system layers, five territorial scales, and four impact pillars: environmental, social, economic, and governance. Its value lies in making local interventions measurable without disconnecting them from the broader systems they seek to influence. The framework becomes operational when implementation starts at community scale, focuses on a feasible core of indicators, and embeds data collection in routine organisational practice. The framework draws on systems thinking, multiscalar sustainability assessment, life cycle analysis, material flow accounting, circular economy research, and governance studies to ensure that indicators remain analytically coherent across scales and dimensions. An actionable methodology follows through five stages: boundary definition, activity mapping, indicator selection, data protocol design, and review. The framework further distinguishes between high-feasibility indicators, which can be derived from records labs already maintain, and lower-feasibility indicators that require dedicated infrastructure and should be deferred to later implementation phases. When implemented through this staged methodology, it supports rigorous measurement, organisational learning, accountability, and comparability across the Fab Lab and Fab City network.

Keywords: Fab City, metrics, systems thinking, circular economy, impact

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